Quantum Mechanics II — Exercise Sheet 0

Operators and their domains

Variant: Solutions

Exercise 1 (A domain changes the adjoint). On $\ell^2(\mathbb{N})$ let A act by $(Ac)_n = n^2 c_n$ on $D(A) = \{c : \sum_{n\geq 1} n^4 |c_n|^2 < \infty\}$. Let B have the same action on $D(B) = \{c \in D(A) : c_1 = 0\}$. (i) Decide which operator is densely defined. For the densely defined operator, determine its adjoint and decide whether it is self-adjoint. (ii) Explain why the lecture's definition of an adjoint does not apply to the other operator.

Solution. The finite sequences belong to $D(A)$ and are dense, so A is densely defined. The adjoint condition is $\langle Ac, d\rangle = \langle c, e\rangle$ for every $c \in D(A)$ and some $e \in \ell^2$. Testing c on the nth coordinate gives

$$e_n = n^2 d_n, \qquad \sum_{n\geq 1} |e_n|^2 = \sum_{n\geq 1} n^4 |d_n|^2.$$

Thus $D(A^*) \subseteq D(A)$. Conversely, for $d \in D(A)$ the required equality follows by Cauchy–Schwarz, with $e = Ad$. Hence $D(A^*) = D(A)$ and $A^* = A$. The closure of $D(B)$ is the proper closed subspace $\{c : c_1 = 0\}$. It is not dense in ℓ^2. An adjoint vector for B would not be unique: adding any multiple of the first coordinate vector leaves all its pairings with $D(B)$ unchanged.

Exercise 2 (A bounded function of position). On $L^2(\mathbb{R})$ consider multiplication by $g(x) = \arctan x$, on its maximal domain. (i) Determine the domain and the operator norm. (ii) Decide whether the operator is self-adjoint and whether it has eigenvalues. Use only Lecture 0, without a claim about its spectrum.

Solution. Since $|\arctan x| \leq \pi/2$, every square-integrable function belongs to the domain. The multiplication lemma gives self-adjointness. The same bound gives $\|M_g\| \leq \pi/2$. For $R > 0$ the unit vector $f_R = \mathbf{1}_{[R,R+1]}$ satisfies

$$\|M_g f_R\| \geq \arctan R \longrightarrow \frac{\pi}{2}.$$

Therefore $\|M_g\| = \pi/2$. Each value of g is taken at at most one point, so the multiplication lemma excludes eigenvalues.

Exercise 3 (Two multiplication domains). Let Q be multiplication by x on its maximal domain in $L^2(\mathbb{R})$. For $a > 1/4$ let $f_a(x) = (1 + x^2)^{-a}$. Determine exactly when $f_a \in D(Q)$ and when $f_a \in D(Q^2)$. Include the threshold cases and give a vector in $D(Q) \setminus D(Q^2)$.

Solution. Membership in $D(Q)$ requires the integral of $x^2 |f_a(x)|^2$ to be finite. At infinity the integrands for the two domains have the orders

$$x^2 |f_a(x)|^2 \sim |x|^{2-4a}, \qquad x^4 |f_a(x)|^2 \sim |x|^{4-4a}.$$

There is no singularity at zero. An integral of x^b on $[1, \infty)$ converges exactly when $b < -1$. Consequently

$$f_a \in D(Q) \iff a > \frac{3}{4}, \qquad f_a \in D(Q^2) \iff a > \frac{5}{4}.$$

At equality the corresponding integral diverges logarithmically. The vector f_1 belongs to $D(Q)$ but not to $D(Q^2)$.

Exercise 4 (Unitary equivalence includes the domain). On $L^2(\mathbb{R})$ let $(Jf)(x) = f(-x)$ and let M_+ be multiplication by e^x on its maximal domain. Find $J^{-1}M_+J$, including its domain. Check whether $f(x) = e^{-|x|}$ belongs to that domain.

Solution. Changing the variable x to $-x$ in the norm shows that J is unitary and $J^{-1} = J$. The transformed operator acts by multiplication by e^{-x}. The unitary-equivalence lemma specifies its domain as

$$D(J^{-1}M_+J) = \{f : e^{-x}f(x) \in L^2(\mathbb{R})\}.$$

For the proposed f, the product $e^{-x}f(x)$ equals 1 on the negative half-line. It is not square-integrable there, so f does not belong to the domain.